



Volumes of Revolution · Mark Scheme

Core Pure 1 Chapter 5: Volumes of Revolution. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – About the x -axis

$$1 \quad V = \pi \int_0^2 9x^2 dx = \pi [3x^3]_0^2 \text{ M1}; = 24\pi \text{ A1.}$$

$$2 \quad V = \pi \int_0^1 (x^4 + 2x^2 + 1) dx \text{ M1}; = \pi \left[\frac{1}{5}x^5 + \frac{2}{3}x^3 + x \right]_0^1 \text{ A1}; = \frac{28\pi}{15} \text{ A1.}$$

$$3 \quad V = \pi \int_1^4 x^{-2} dx \text{ M1}; = \pi [-x^{-1}]_1^4 \text{ A1}; = \pi \left(1 - \frac{1}{4}\right) = \frac{3\pi}{4} \text{ A1.}$$

$$4 \quad \text{Limits } x = 0 \text{ and } x = 4 \text{ B1}; V = \pi \int_0^4 (16x^2 - 8x^3 + x^4) dx \text{ M1}; = \pi \left[\frac{16}{3}x^3 - 2x^4 + \frac{1}{5}x^5 \right]_0^4 \text{ A1}; = \frac{512\pi}{15} \text{ A1.}$$

$$5(a) \quad \frac{dy}{dx} = 1 - \frac{1}{x^2} = 0 \Rightarrow x = 1 \text{ M1}; \text{ minimum point } (1, 2) \text{ A1.}$$

$$5(b) \quad y^2 = x^2 + 2 + x^{-2} \text{ B1}; V = \pi \int_1^3 (x^2 + 2 + x^{-2}) dx \text{ M1}; = \pi \left[\frac{1}{3}x^3 + 2x - x^{-1} \right]_1^3 \text{ A1}; = \pi \left(\frac{44}{3} - \frac{4}{3} \right) = \frac{40\pi}{3} \text{ A1.}$$

5(c) 8π is the volume of the cylinder formed by rotating the rectangle $1 \leq x \leq 3$, $0 \leq y \leq 2$ (radius 2, length 2) B1; since the minimum of C is $y = 2$, the curve lies on or above $y = 2$ for $1 \leq x \leq 3$, so the solid contains this cylinder B1.

Section B – About the y -axis

$$6 \quad V = \pi \int_0^4 y dy \text{ M1}; = 8\pi \text{ A1.}$$

$$7 \quad x^2 = y^{2/3}, V = \pi \int_1^8 y^{2/3} dy \text{ M1}; = \pi \left[\frac{3}{5}y^{5/3} \right]_1^8 \text{ A1}; = \frac{3}{5}(32 - 1)\pi = \frac{93\pi}{5} \text{ A1.}$$

$$8 \quad x = e^y, \text{ so } V = \pi \int_0^2 e^{2y} dy \text{ M1}; = \pi \left[\frac{1}{2}e^{2y} \right]_0^2 \text{ A1}; = \frac{\pi}{2}(e^4 - 1) \text{ A1.}$$

$$9 \quad x = \frac{2}{y} - 1; \text{ limits } y = 1 \text{ to } y = 2 \text{ B1}; V = \pi \int_1^2 \left(\frac{4}{y^2} - \frac{4}{y} + 1 \right) dy \text{ M1}; = \pi \left[-\frac{4}{y} - 4 \ln y + y \right]_1^2 \text{ A1}; = \pi((-4 \ln 2) - (-3)) = \pi(3 - 4 \ln 2) \text{ A1.}$$

$$10(a) \quad (4, 0) \text{ B1}; (0, 2) \text{ B1.}$$

$$10(b) \quad V = \pi \int_0^2 (4 - y^2)^2 dy \text{ M1}; = \pi \int_0^2 (16 - 8y^2 + y^4) dy \text{ A1}; = \pi \left[16y - \frac{8}{3}y^3 + \frac{1}{5}y^5 \right]_0^2 \text{ M1}; = \frac{256\pi}{15} \text{ A1.}$$

$$10(c) \quad y^2 = 4 - x, V = \pi \int_0^4 (4 - x) dx \text{ M1}; = 8\pi \text{ A1.}$$

Section C – Subtraction and modelling

$$11 \quad \text{Cylinder minus cone: } V = \pi \int_0^2 (4 - x^2) dx \text{ M1}; = \pi \left(8 - \frac{8}{3}\right) = \frac{16\pi}{3} \text{ A1.}$$

$$12 \quad \text{Limits } x = 0 \text{ and } x = 1 \text{ B1}; V = \pi \int_0^1 (x^2 - x^4) dx \text{ M1}; = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15} \text{ A1.}$$

$$13 \quad \text{Meet at } (0, 0) \text{ and } (2, 4); \text{ outer } x^2 = y, \text{ inner } x^2 = \frac{1}{4}y^2 \text{ B1}; V = \pi \int_0^4 \left(y - \frac{1}{4}y^2\right) dy \text{ M1}; = \pi \left(8 - \frac{16}{3}\right) = \frac{8\pi}{3} \text{ A1.}$$

$$14 \quad \text{Limits } x = \pm 1 \text{ B1}; V = \pi \int_{-1}^1 ((4 - x^2)^2 - 3^2) dx \text{ M1}; = \pi \int_{-1}^1 (7 - 8x^2 + x^4) dx = \pi \left[7x - \frac{8}{3}x^3 + \frac{1}{5}x^5 \right]_{-1}^1 \text{ A1}; = \frac{136\pi}{15} \text{ A1.}$$



15(a) Outer solid: $\pi \int_0^6 5y \, dy = 90\pi$ M1 A1; hollow: $\pi \int_1^6 5(y-1) \, dy = \frac{125\pi}{2}$ M1; $90\pi - \frac{125\pi}{2} = \frac{55\pi}{2}$ A1.

15(b) Water from $y = 1$ to $y = 4$: $\pi \int_1^4 5(y-1) \, dy$ M1; $= \frac{45\pi}{2} = 70.7 \text{ cm}^3$ A1.

15(c) Mass $= 2.4 \times \frac{55\pi}{2} = 66\pi = 207 \text{ g}$ B1.

15(d) E.g. the bowl has no flat base so would not stand level; the rim is not a real shape (it has a sharp edge); the material is assumed uniform B1.

Roar-level challenge

16 Outer radius $y_1 = 3 + \sqrt{1-x^2}$, inner radius $y_2 = 3 - \sqrt{1-x^2}$ M1; $y_1^2 - y_2^2 = 12\sqrt{1-x^2}$ A1; $V = \pi \int_{-1}^1 12\sqrt{1-x^2} \, dx$ M1; $\int_{-1}^1 \sqrt{1-x^2} \, dx = \frac{\pi}{2}$ (area of a semicircle of radius 1, or substitute $x = \sin \theta$) A1; $V = 12\pi \times \frac{\pi}{2} = 6\pi^2$ A1.